

A Smooth Strongly Convex Obstruction to Complexity-Free Uniform Subdifferential Convergence

Verification note: partial progress on a question of Ruan

Abstract

We give a self-contained verification of a two-piece max-of-smooth counterexample to uniform Hausdorff convergence of empirical subdifferentials. In dimension two, both smooth pieces are globally C^∞ , have a common deterministic gradient-Lipschitz constant, and are uniformly strongly convex. Nevertheless, for one fixed data distribution, almost surely and simultaneously for every sample size m and every neighborhood of the origin, the Hausdorff distance between the population and empirical regular subdifferentials is at least $1/2$ at some point in that neighborhood.

This result is *partial progress, not a solution* to Ruan’s question about extending his convex-composite theorem to max-of-smooth models. The construction is itself already a scalar stochastic convex-composite model of the form $t_+ \circ c + R$, and its threshold class has infinite VC dimension. It therefore rules out only complexity-free extensions based on smoothness, dimension, the number of pieces, and strong convexity. It does not rule out an extension carrying an explicit active-set complexity term. The core obstruction also appears in the earlier two-piece construction of Tian and Royset; the version proved here strengthens the regularity to C^∞ , imposes strong convexity, and yields an exact obstruction at every sample size.

Status statement 1. *The mathematical counterexample proved below is valid. Its correct classification relative to Ruan’s paper is **partial progress**: it disproves a complexity-free max-of-smooth analogue, but it does not settle whether Theorem 5 of [1] admits a max-of-smooth extension under finite active-set complexity or comparable structural assumptions.*

1 Question, contribution, and scope

Ruan studies stochastic convex-composite objectives of the form

$$\varphi(x) = \mathbb{E}[h(c(x; \xi))] + R(x) + \iota_X(x), \quad \varphi_m(x) = \frac{1}{m} \sum_{i=1}^m h(c(x; \xi_i)) + R(x) + \iota_X(x), \quad (1)$$

where $h : \mathbb{R} \rightarrow \mathbb{R}$ is deterministic, convex, and univariate, while $c(\cdot; \xi)$ is smooth. His Theorem 5 gives a high-probability bound for

$$\sup_{x \in X \cap \mathbb{B}(x_0, r)} H(\widehat{\partial}\varphi(x), \widehat{\partial}\varphi_m(x))$$

whose nontrivial part depends on the VC dimension of the threshold family

$$\mathcal{F}_c = \{\{\xi : c(x; \xi) \geq t\} : x \in \mathbb{R}^d, t \in \mathbb{R}\}; \quad (2)$$

see [1, Theorem 5 and (4.12)]. In the discussion section, Ruan asks whether that theorem can be extended to other weakly convex models, including max-of-smooth functions.

The theorem below proves that no such extension can be based only on the ambient dimension, the number of smooth pieces, deterministic smoothness bounds, and a positive strong-convexity modulus. The obstruction is an active-set family capable of encoding infinitely many independent binary coordinates.

This does not constitute a complete negative answer to Ruan's question. Indeed, the example has the representation

$$\ell(\xi, x) = (c(x; \xi))_+ + R(x),$$

so it already lies in the scalar convex-composite class in (1). Its threshold family (2) has infinite VC dimension, making the VC-dependent estimate in Ruan's theorem vacuous rather than false. Thus a max-of-smooth extension with a finite active-set complexity assumption remains untouched.

Tian and Royset [2] independently give a bivariate convex two-piece obstruction and explicitly use it to show that Ruan's VC-major condition cannot in general be deleted. Their theorem is stronger in its robustness to certain shrinking neighborhood enlargements. The construction below is a closely related variant with different sharpenings: both pieces are globally C^∞ , every sample loss is strongly convex, and the gap is obtained almost surely for every sample size rather than only through an asymptotic lower limit.

2 Notation

For nonempty compact sets $A, B \subset \mathbb{R}^d$, their Hausdorff distance is

$$H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} \|a - b\|, \sup_{b \in B} \inf_{a \in A} \|b - a\| \right\}.$$

We write $\widehat{\partial}f(x)$ for the Fréchet, or regular, subdifferential. At a point where f is differentiable, $\widehat{\partial}f(x) = \{\nabla f(x)\}$. The open Euclidean ball of radius r around x is denoted by $\mathbb{B}(x, r)$.

A family $\mathcal{A}$ of subsets of a set Ξ shatters a finite set $S \subset \Xi$ if $\{A \cap S : A \in \mathcal{A}\} = 2^S$. Its VC dimension $\text{vc}(\mathcal{A})$ is the supremum of the cardinalities of shattered finite sets.

3 Independent binary coordinates

Let $\Xi = [0, 1]$ with Lebesgue probability measure. For $\xi \in [0, 1)$ and $k \geq 1$, define

$$b_k(\xi) = \lfloor 2^k \xi \rfloor - 2 \lfloor 2^{k-1} \xi \rfloor \in \{0, 1\}, \quad \sigma_k(\xi) = 2b_k(\xi) - 1 \in \{-1, 1\}, \quad (3)$$

and set $b_k(1) = 0$.

Lemma 2. *If ξ is uniform on $[0, 1]$, then $(b_k(\xi))_{k \geq 1}$ is an independent sequence of Bernoulli(1/2) random variables. If $\xi_1, \xi_2, \dots$ are independent uniform random variables, then the entire array $\{b_k(\xi_i) : i, k \geq 1\}$ is independent.*

Proof. Fix $n \geq 1$ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{0, 1\}^n$. Up to endpoints of measure zero, the event

$$\{b_1(\xi) = \varepsilon_1, \dots, b_n(\xi) = \varepsilon_n\}$$

is one dyadic interval of length 2^{-n} . Hence its probability is 2^{-n} , proving independence and the Bernoulli(1/2) marginals. Independence across the sample index i follows from the independence of the variables ξ_i . $\square$

4 A flat family of disjoint smooth plateaus

Choose $\rho \in C_c^\infty(\mathbb{R})$ such that

$$0 \leq \rho \leq 1, \quad \text{supp}(\rho) \subset (-1, 1), \quad \rho(t) = 1 \quad \text{for } |t| \leq \frac{1}{2}. \quad (4)$$

For $k \geq 1$, put

$$c_k = 2^{-k}, \quad s_k = \frac{c_k}{8}, \quad a_k = \exp\left(-\frac{1}{s_k^2}\right), \quad (5)$$

and define

$$\psi_k(v) = a_k \rho\left(\frac{v - c_k}{s_k}\right), \quad v \in \mathbb{R}. \quad (6)$$

Then

$$\text{supp}(\psi_k) \subset I_k := [c_k - s_k, c_k + s_k] = [7s_k, 9s_k].$$

The intervals I_k are pairwise disjoint because

$$c_{k+1} + s_{k+1} = \frac{9}{16}c_k < \frac{7}{8}c_k = c_k - s_k.$$

For $\xi \in \Xi$, set

$$q_\xi(v) = \sum_{k=1}^{\infty} \sigma_k(\xi) \psi_k(v). \quad (7)$$

At each $v \neq 0$, at most one summand is nonzero.

Lemma 3 (Uniform C^∞ regularity and flatness). *For every $\xi \in \Xi$, the function q_ξ belongs to $C^\infty(\mathbb{R})$ and*

$$q_\xi^{(p)}(0) = 0 \quad \text{for every integer } p \geq 0.$$

Moreover, for every $p \geq 0$ there is a finite deterministic constant M_p such that

$$\sup_{\xi \in \Xi} \sup_{v \in \mathbb{R}} |q_\xi^{(p)}(v)| \leq M_p. \quad (8)$$

More precisely, for every $p, N \geq 0$,

$$\sup_{\xi \in \Xi} |q_\xi^{(p)}(v)| = o(|v|^N) \quad \text{as } v \rightarrow 0. \quad (9)$$

Proof. Away from zero, the assertion follows from the disjoint supports and the smoothness of each ψ_k . If $v \in I_k$, then

$$|\psi_k^{(p)}(v)| \leq \|\rho^{(p)}\|_\infty e^{-1/s_k^2} s_k^{-p}. \quad (10)$$

Since $7s_k \leq |v| \leq 9s_k$ on I_k , for all $p, N \geq 0$,

$$\sup_{v \in I_k} \frac{|\psi_k^{(p)}(v)|}{|v|^N} \leq 7^{-N} \|\rho^{(p)}\|_\infty e^{-1/s_k^2} s_k^{-(p+N)} \longrightarrow 0. \quad (11)$$

The convergence follows because e^{-1/s^2} decays faster than every power as $s \downarrow 0$.

Extend the piecewise derivative $q_\xi^{(p)}$ by zero at the origin. Estimate (11) gives continuity there and, with $N = 1$,

$$\frac{q_\xi^{(p)}(v) - q_\xi^{(p)}(0)}{v} \longrightarrow 0.$$

Induction on p shows that all derivatives exist at zero and equal zero. The same estimate proves (9), uniformly in ξ . Finally, at most one bump is active at any point, and

$$\sup_{k \geq 1} e^{-1/s_k^2} s_k^{-p} < \infty;$$

combining this with (10) proves (8). $\square$

Set

$$L := \sup_{\xi \in \Xi, v \in \mathbb{R}} |q_\xi''(v)| < \infty, \quad \lambda = L + 1. \quad (12)$$

For $x = (u, v) \in \mathbb{R}^2$, define

$$c(x; \xi) = u + q_\xi(v), \quad (13)$$

$$h_0(x) = \frac{\lambda}{2} \|x\|^2, \quad h_1(\xi, x) = c(x; \xi) + \frac{\lambda}{2} \|x\|^2, \quad (14)$$

and

$$\ell(\xi, x) = \max\{h_0(x), h_1(\xi, x)\} = (c(x; \xi))_+ + \frac{\lambda}{2} \|x\|^2. \quad (15)$$

5 The counterexample

Theorem 4 (Exact obstruction at every sample size). *The loss in (15) has the following properties.*

- (i) *For every ξ , the functions h_0 and $h_1(\xi, \cdot)$ are globally C^∞ and 1-strongly convex. Their gradients have a common deterministic global Lipschitz constant. Consequently, $\ell(\xi, \cdot)$ is 1-strongly convex.*
- (ii) *The map $\ell : \Xi \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is Carathéodory and $\mathbb{E}|\ell(\xi, x)| < \infty$ for every x .*
- (iii) *Let $\xi_1, \xi_2, \dots$ be iid uniform random variables and define*

$$\Phi(x) = \mathbb{E}[\ell(\xi, x)], \quad \Phi_m(x) = \frac{1}{m} \sum_{i=1}^m \ell(\xi_i, x). \quad (16)$$

Then, with probability one, simultaneously for every $m \geq 1$ and every $r > 0$,

$$\sup_{x \in \mathbb{B}(0, r)} H(\widehat{\partial}\Phi(x), \widehat{\partial}\Phi_m(x)) \geq \frac{1}{2}. \quad (17)$$

Proof. Step 1: smoothness and strong convexity. By Lemma 3, both pieces are globally C^∞ . Their Hessians are

$$\nabla^2 h_0(x) = \lambda I_2, \quad \nabla^2 h_1(\xi, x) = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda + q_\xi''(v) \end{pmatrix}.$$

Since $|q_\xi''(v)| \leq L$ and $\lambda = L + 1$, both Hessians dominate I_2 . Thus the pieces are 1-strongly convex. Their Hessian operator norms are bounded, for example, by 2λ , so their gradients have a common deterministic global Lipschitz constant. The maximum of functions sharing a strong-convexity modulus has the same modulus: indeed,

$$\ell(\xi, x) - \frac{1}{2} \|x\|^2 = \max_{j \in \{0, 1\}} \left(h_j(\xi, x) - \frac{1}{2} \|x\|^2 \right)$$

is convex.

Step 2: measurability and integrability. For fixed $x = (u, v)$, at most one binary coordinate $b_k(\xi)$ enters $q_\xi(v)$, so $\xi \mapsto \ell(\xi, x)$ is measurable. For fixed ξ , the map $x \mapsto \ell(\xi, x)$ is continuous. Thus ℓ is Carathéodory. Since at most one bump is active,

$$|q_\xi(v)| \leq A := \sup_{k \geq 1} a_k < \infty,$$

and hence

$$0 \leq \ell(\xi, x) \leq |u| + A + \frac{\lambda}{2} \|x\|^2.$$

This proves integrability.

Step 3: a local deterministic gap. Let

$$x_k = (0, c_k) \tag{18}$$

and define the open rectangle

$$U_k = \left\{ (u, v) : |u| < \frac{a_k}{2}, |v - c_k| < \frac{s_k}{2} \right\}. \tag{19}$$

On U_k , the plateau property of ρ and the disjointness of the supports give

$$q_\xi(v) = \sigma_k(\xi) a_k.$$

Therefore, for $x = (u, v) \in U_k$,

$$(c(x; \xi))_+ = \begin{cases} u + a_k, & b_k(\xi) = 1, \\ 0, & b_k(\xi) = 0. \end{cases} \tag{20}$$

Indeed, $u + a_k > a_k/2 > 0$ and $u - a_k < -a_k/2 < 0$.

Taking expectation in (20), the population objective is smooth throughout U_k and satisfies

$$\Phi(u, v) = \frac{1}{2}(u + a_k) + \frac{\lambda}{2}(u^2 + v^2). \tag{21}$$

Consequently,

$$\widehat{\partial}\Phi(x_k) = \left\{ \left(\frac{1}{2}, \lambda c_k \right) \right\}. \tag{22}$$

Fix m . If

$$b_k(\xi_1) = \dots = b_k(\xi_m) = 1, \tag{23}$$

then every sample loss uses the active branch throughout U_k , and hence

$$\Phi_m(u, v) = u + a_k + \frac{\lambda}{2}(u^2 + v^2) \quad ((u, v) \in U_k). \tag{24}$$

It follows that

$$\widehat{\partial}\Phi_m(x_k) = \{(1, \lambda c_k)\}. \tag{25}$$

Since the Hausdorff distance between two singleton sets is the distance between their elements, (22) and (25) yield

$$H(\widehat{\partial}\Phi(x_k), \widehat{\partial}\Phi_m(x_k)) = \frac{1}{2}. \tag{26}$$

Step 4: a favorable coordinate exists infinitely often. For fixed m and k , define

$$A_{m,k} = \{b_k(\xi_1) = \dots = b_k(\xi_m) = 1\}.$$

By Lemma 2, the events $(A_{m,k})_{k \geq 1}$ are independent and $\mathbb{P}(A_{m,k}) = 2^{-m}$. Thus

$$\sum_{k=1}^{\infty} \mathbb{P}(A_{m,k}) = \infty,$$

and the second Borel–Cantelli lemma implies that $A_{m,k}$ occurs for infinitely many k , almost surely. Taking the countable intersection over $m \geq 1$, we obtain an event of probability one on which this is true simultaneously for every m .

On that event, fix $m \geq 1$ and $r > 0$. Because $x_k = (0, 2^{-k}) \rightarrow 0$ and favorable indices k are unbounded, there is a favorable k with $x_k \in \mathbb{B}(0, r)$. Equation (26) then proves (17). $\square$

Corollary 5 (No complexity-free vanishing rate). *For the single model in Theorem 4, let $b_m \downarrow 0$ be arbitrary and fix $\delta \in (0, 1)$. For all sufficiently large m ,*

$$\mathbb{P} \left(\sup_{x \in \mathbb{B}(0, r)} H(\hat{\partial}\Phi(x), \hat{\partial}\Phi_m(x)) \leq b_m \right) = 0 \quad \text{for every } r > 0.$$

In particular, no uniform theorem with a rate tending to zero can hold over all two-piece max-of-smooth models using only fixed bounds on dimension, number of pieces, gradient Lipschitz constants, and strong convexity.

Proof. By Theorem 4, the displayed supremum is at least $1/2$ almost surely for every m . For sufficiently large m , $b_m < 1/2$. $\square$

6 The active-set class has infinite VC dimension

The obstruction can be identified exactly at the level of the threshold sets used in Ruan’s analysis.

Proposition 6 (Infinite threshold complexity). *For the inner function c in (13), let*

$$\mathcal{F}_c = \{ \{ \xi \in [0, 1] : c(x; \xi) \geq t \} : x \in \mathbb{R}^2, t \in \mathbb{R} \}.$$

Then $\text{vc}(\mathcal{F}_c) = \infty$.

Proof. For $k \geq 1$, define the coordinate set

$$E_k = \{ \xi : b_k(\xi) = 1 \}.$$

At $x_k = (0, c_k)$, one has $c(x_k; \xi) = \sigma_k(\xi)a_k$, and hence

$$E_k = \{ \xi : c(x_k; \xi) \geq 0 \} \in \mathcal{F}_c. \tag{27}$$

It is therefore enough to show that the family $\{E_k : k \geq 1\}$ has infinite VC dimension.

Fix $N \geq 1$, and enumerate the 2^N vectors in $\{0, 1\}^N$ as $y^1, \dots, y^{2^N}$, where $y^k = (y_1^k, \dots, y_N^k)$. For $i = 1, \dots, N$, define

$$\zeta_i = \sum_{k=1}^{2^N} y_i^k 2^{-k} \in [0, 1).$$

The points $\zeta_1, \dots, \zeta_N$ are distinct: if $i \neq j$, one of the enumerated vectors has i th coordinate 1 and j th coordinate 0, so the corresponding binary digits of ζ_i and ζ_j differ. By the definition (3), $b_k(\zeta_i) = y_i^k$ for $1 \leq k \leq 2^N$. Hence, for every subset $S \subset \{1, \dots, N\}$, there is an index k for which $y_i^k = \mathbf{1}\{i \in S\}$ for all i . For that k ,

$$E_k \cap \{\zeta_1, \dots, \zeta_N\} = \{\zeta_i : i \in S\}.$$

Thus $\{\zeta_1, \dots, \zeta_N\}$ is shattered. Since N was arbitrary, $\text{vc}(\mathcal{F}_c) = \infty$. $\square$

Remark 7 (The empirical-process mechanism). *The proof of Theorem 4 is the same obstruction in empirical-process language. On the favorable event (23),*

$$\mathbb{P}(E_k) = \frac{1}{2}, \quad \frac{1}{m} \sum_{i=1}^m \mathbf{1}_{E_k}(\xi_i) = 1.$$

Thus the empirical frequency of a sample-dependent active set differs from its population probability by 1/2. Smoothness in the decision variable does not control this combinatorial freedom in the data variable.

7 Precise relation to Ruan's theorem

Proposition 8 (Why the construction is only partial progress). *The counterexample has all of the following properties.*

- (a) *It is already a stochastic convex-composite model of the precise scalar form considered in [1, Section 4]: take*

$$h(t) = t_+, \quad c((u, v); \xi) = u + q_\xi(v), \quad R(u, v) = \frac{\lambda}{2}(u^2 + v^2), \quad X = \mathbb{R}^2.$$

- (b) *The analytic smoothness and tail assumptions entering Ruan's Theorem 5 hold on every bounded ball with finite deterministic constants.*
- (c) *The threshold family governing Ruan's bound has infinite VC dimension, by Proposition 6. Consequently, the VC-dependent rate in that theorem is vacuous for this example and is not contradicted.*

Therefore the example excludes a complexity-free extension, but does not exclude a max-of-smooth extension under finite VC dimension, finite entropy, definability, a margin condition, or another quantitative active-set restriction.

Proof. Part (a) is the identity (15). For part (b), the gradient

$$\nabla_x c((u, v); \xi) = (1, q'_\xi(v))$$

is uniformly bounded by Lemma 3, and

$$\|\nabla c(x; \xi) - \nabla c(y; \xi)\| \leq L\|x - y\|$$

holds deterministically. For $h(t) = t_+$, one may take the smooth component in Ruan's decomposition to be zero, so the associated smooth term $e(x, \xi)$ vanishes. Bounded random vectors are subexponential, and the deterministic Lipschitz estimate supplies the required increment and integrable-Lipschitz bounds. Thus the analytic conditions are satisfied with finite constants. Part (c) is Proposition 6; Ruan's rate contains $\text{vc}(\mathcal{F}_c)$, so it gives no finite estimate here. $\square$

Remark 9 (What would constitute a genuine solution?). *A full positive solution would give a max-of-smooth analogue of Ruan’s theorem under explicit assumptions controlling the active regions and the gradients of the active pieces. A full negative solution would have to defeat every natural counterpart of the complexity control already present in Ruan’s theorem, or show that no such counterpart can yield the desired conclusion. The present example does neither. It establishes the narrower but useful necessity of some active-set complexity or stability hypothesis.*

8 Comparison with Tian–Royset

Tian and Royset [2, Theorem 3 and the subsequent discussion] construct a random convex function in dimension two of the form

$$f(\xi, x) = \max\{g(\xi, x), 0\} + 35\|x\|^2,$$

where $g(\xi, \cdot)$ is smooth with a uniform Lipschitz-gradient bound. Their construction also encodes the binary digits of a uniform random variable in disjoint plateaus, and they explicitly note that it belongs to Ruan’s scalar convex-composite class with an infinite-VC threshold family. They conclude that the VC-major assumption cannot generally be dropped, even when an explicit distribution-free rate is not requested.

Accordingly, the central mathematical message of the present note is not new. The exact version in Theorem 4 nevertheless records three clean strengthenings of that mechanism:

- (i) the two component functions are globally C^∞ rather than merely C^1 with Lipschitz gradient;
- (ii) both components, and hence every sample loss, are uniformly strongly convex;
- (iii) the lower bound holds almost surely for every sample size and inside every neighborhood of a fixed point.

Tian and Royset obtain other stronger conclusions, notably robustness under shrinking spatial enlargements. The results should therefore be viewed as complementary formulations of the same infinite-complexity obstruction, not as competing resolutions of Ruan’s broader max-of-smooth question.

9 Conclusion

The counterexample is mathematically correct and shows that even global C^∞ smoothness of two branches, deterministic smoothness constants, fixed dimension, and strong convexity do not force uniform convergence of empirical subdifferentials. The decisive failure is combinatorial: the active-set class can encode infinitely many independent bits.

The proper conclusion relative to Ruan’s paper is therefore:

A complexity-free extension of Theorem 5 to arbitrary max-of-smooth models is impossible. This is partial progress, not a complete solution, because the example has infinite active-set complexity and already lies in Ruan’s existing scalar convex-composite framework.

References

- [1] F. Ruan, On the uniform convergence of subdifferentials in stochastic optimization and learning, arXiv:2405.10289, version 6, 2025.

- [2] L. Tian and J. O. Royset, Failure of uniform laws of large numbers for subdifferentials and beyond, arXiv:2511.16568, version 2, 2026.